Every finite characteristic-two field: trace and geometric torsion

Result. For every finite field of characteristic two and every nonsingular general Weierstrass model, Lean proves the exact source gcd(q+1−#E,q)≠1 predicate iff a1=0 iff j=0 iff odd actual point count iff rational two-torsion cardinality one iff geometric two-torsion cardinality one. No field-size or extension-degree bound, Hasse, trace-square classification or endomorphism-rank premise. [pcc:unverified]

Every finite char2	Source gcd(trace,q)≠1 ↔ a1=0 ↔ j=0 ↔ odd #E ↔ rational/geometric two-torsion count 1
Drop finiteness	F2(X): $y^2+xy=x^3+X$, Δ=X≠0, j≠0 Rational two-torsion count 1; geometric count 2
Change to char3	Credited F3 curve: actual #E=5 Rational three-torsion count 1; source predicate false

Actual finite-field bridge. If a1≠0, a nonzero point killed by two has x=−a3/a1. Its equation becomes a square-root equation; the official finite reduced-ring square theorem supplies its ordinate. Actual Cauchy and Lagrange give even point count iff a1≠0. Official FiniteField.card supplies $q=2^n$, n>0; coprime-power/parity arithmetic connects this to the source trace predicate. The preceding actual geometric j theorem supplies the final geometric interpretation. [pcc:unverified]

Modules and scalar extensions. Actual rational two-torsion has cardinality one/two and induced ZMod2 rank zero/one according as j=0/nonzero. Its cardinality equals geometric two-torsion cardinality over every algebraically closed scalar extension. The source predicate is invariant under every finite field extension, using the actual scalar j identity. Prior group, module, Cauchy, geometric j and finite-field perfectness proofs are credited. [pcc:unverified]

Negative controls. Over F2(X), $y^2+xy=x^3+X$ has Δ=X≠0, a1=1 and j≠0. Rational two-torsion count is one, geometric count two. X is not a square because a square has even integer degree; Lean directly negates ordinary rational descent after deleting finiteness. The credited F3 curve $y^2=x^3+2x^2+1$ has five actual rational points/rational three-torsion count one/source predicate false; Lean directly negates changing characteristic two to three in the rational criterion. These test hypotheses, not qualified-source theorems. [pcc:unverified]

Primary review. FST2010 selected pp230/237: exact trace predicate and embedding qualifications. Milne Remark3.16/context pp98–99 and p75 notes: geometric p-rank versus endomorphism-rank distinction; prior PDF104/80 inspections credited. Official Mathlib finite reduced-ring squares, finite-field cardinality, rational-function degree and j identities read. KB/Conductor unavailable; bounded fallback preceded code. No full-paper/book read, novelty or corpus promotion. [pcc:unverified]

Validation and scope. Whole gate 26 declarations, zero failures, zero trusted native exceptions. All 65 build/provenance files retained losslessly in 26 objects; five gate files, complete statement and recursive source/config pins. Lean4.35.0/Mathlib56a00585…, standard foundations only. One unsuppressed legacy/modern-module compatibility advisory remains; attempted module migration failed against frozen legacy imports and was reverted. Exact local replay differs from signed standing. General characteristic-three/other-prime trace-geometric and endomorphism-rank identifications, Hasse/embedding bounds and remaining original fourteen-item gaps stay open. No qualified-source refutation, holes/custom axioms/native evaluator/raised limits/new global flags/waivers/native experiment/paid compute/agent upload. [pcc:unverified]